Tutorial Limits, Continuity and Asymptotes

1. Find the following limits, if they exists. If not explain why:

(a) $\lim_{x\to -1} \frac{3x^3-2x^2+1}{4x^2+2x-3}$,

(b) $\lim_{x\to -1} \frac{2x^2-10x-12\ln(\ln(e^e))}{7\sqrt{x+10}-21}$,

(c) $\lim_{x\to 0^+} \frac{2}{1-e^{\frac{1}{x}}}$,

(d) $\lim_{x\to 0^-} \frac{2}{1-e^{\frac{1}{x}}}$,

(e) $\lim_{x\to 0} \frac{2}{1-e^{\frac{1}{x}}}$,

(f) $\lim_{x\to 0^+} \sqrt{x}\ e^{\sin\frac{\pi}{x}}$

2. Use the $\epsilon-\delta$-definition of limit to prove that:

$$\lim_{x\to -2} 2x+3=-1,$$

3. Find $M>0$, as smallest as possible for you, such that if $|x-2|<1$; then, $|x-4|<M$.
Answer: $M=7$ (or any other number greater than 7).

4. Prove, using the $(\epsilon-\delta)$-definition of limit that

$$\lim_{x\to 2} x^2-2x-7=1$$

5. Find all possible paramters $a,b\in\mathbb{R}$, such that it is possible to define f at 0, such that f is continuous at $x=0$.

$$f(x)=\begin{cases} ab\ \frac{\cos x-1}{x^2} & \text{if } x<0,\\ \frac{\sqrt{ax+25}-5}{\sqrt{x+5}=\sqrt{5}} & \text{if } x>0 \end{cases} \tag{1}$$

6. For which values of $c\in\mathbb{R}$ is the function f continuous on $\mathbb{R}$?. For which values $c\in\mathbb{R}$ does the function f have a vertical asymptote?. Compute the horizontal asympteтes of f.

$$f(x)=\begin{cases} \frac{x^3+7x^2+cx-40}{x+4} & \text{if } x<-4,\\ \ln(x^2+10x+25)+x+\ln\frac{1}{e^{x+6}} & \text{if } x\geq 4. \end{cases} \tag{2}$$

7. Consider the functioon $f(x)=\sqrt{x}$. Use the definition of the derivative, to compute the derivative at any point.

8. The figure shows the graphs of f, f' and f''. Identify each curve and explain your choices.

9. Consider the following function:

$$f(x)=\begin{cases} 0\ \frac{\cos x-1}{x^2} & \text{if } x<0,\\ 5-x & \text{if } 0<x<4,\\ \frac{1}{5-x} & \text{if } x\geq 4. \end{cases} \tag{3}$$

(a) Is f differentiable at $x=0$?

(b) Use the **definition** of the **left-hand** and **right-hand derivatives** of f at a, to answer if the following function is differentiable at $x=4$.